-: 12th HACKER JUGAAD NOTES :-

Determination Introduction

A determinant is a particular type of expression written in a special concise form of rows and columns, equal in number.

For example

$\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ is a determinant having 2 rows and 2 columns, hence it is of second order, the number a_1, b_1, a_2, b_2 are called the elements of the determinant.

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Determinant

- If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a matrix then determinant of matrix A is written as $|A|$ or $\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$

- Only square matrices have determinants.

Minor of an element

The determinant obtained by deleting the row and the column in which the given element lies.

$$A = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$M_{11} = a_{22}a_{33} - a_{23}a_{32} = C_{11}$

$M_{32} = a_{11}a_{23} - a_{21}a_{13} = -C_{32}$

$$A = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$M_a = d$ | $C_a = d$

$M_b = c$ | $C_b = -c$

$M_c = b$ | $C_c = -b$

$M_d = a$ | $C_d = a$

Cofactor of an element a_{ij}

Cofactor of the element a_{ij} is denoted by C_{ij} and is given by

$$C_{ij} = (-1)^{i+j} M_{ij}$$

where i and j denotes the row and column in which the particular element lies.

If $i + j$ = even $\Rightarrow C_{ij} = M_{ij}$

If $i + j$ = odd $\Rightarrow C_{ij} = -M_{ij}$

Expansion of a Determinant

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$ → R_1

$= a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23}$ → R_2

along C_3

$= a_{13}C_{13} + a_{23}C_{23} + a_{33}C_{33}$

$a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23} = 0$

$a_{12}C_{13} + a_{22}C_{23} + a_{32}C_{33} = 0$

Note :-

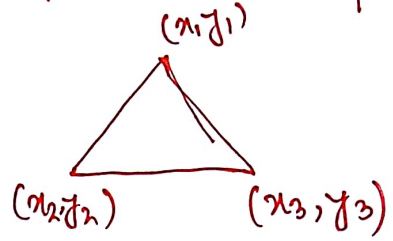
i) The sum of the product of elements of any row (or column) with their corresponding co-factors is always equal to the value of the determinant.

ii) The sum of the product of elements of any row (or column) with the cofactors of other row (or column) is always equal to zero.

Area of a Triangle

The area of the triangle whose vertices are (x_1, y_1), (x_2, y_2) and (x_3, y_3) is

$$= \left| \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \right| \rightarrow \text{mod}$$



i) The area is positive, take only absolute value.

ii) If the three points are collinear, the area of triangle is zero.

Adjoint of Matrix

Let $A = |a_{ij}|$ be a square matrix of order n and let C_{ij} be cofactor of a_{ij} in A then the adjoint of A. denoted of A, denoted by adj A, is defined as the transpose of the cofactor matrix.

$$\text{adj } A = |C_{ij}|^T \Rightarrow \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{23} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}^n$$

Ex → $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\text{adj } A = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}^T$$

$$\text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$ → Change sign, → Inter change

Adjoint of a (2×2) Matrix by Trick:-

Ex → $A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}$

⇒ $\text{adj } A = \begin{bmatrix} 7 & -3 \\ -5 & 1 \end{bmatrix}$

Ex → $A = \begin{bmatrix} 7 & 2 \\ -3 & 8 \end{bmatrix}$

$\text{adj } A = \begin{bmatrix} -8 & -2 \\ 3 & 7 \end{bmatrix}$

Properties of Adjoint of a Matrix :-

Property 1

For any square matrix A of order n, $\boxed{A(\text{adj } A) = (\text{adj } A)A = |A| I_n}$

Property 2

Let A be a non-singular matrix of order n. Then

$\boxed{|\text{adj } A| = |A|^{n-1}}$

For, Ex :- $n = 3$

$|\text{Adj}| = |A|^{3-1}$

$= |A|^2$

Important property

Ex = $|kA| = k^n |A|$ → n is order

Ex = $|3A| = 3^n |A|$

Inverse of a Matrix

A square matrix A said to be **invertible** if and only if there exists a matrix B such that, $\boxed{AB = I = BA}$

B is called the inverse of A and is denoted by $\boxed{A^{-1}}$. Thus $A^{-1} = B$

We know that $A(\text{adj } A) = (\text{adj } A)A = |A| I$

$$\boxed{A = \frac{\text{adj } A}{|A|} = \frac{(\text{adj } A)}{|A|} A = I}$$

$$\boxed{B = \frac{\text{adj } A}{|A|}}$$ → For inverse to exist

$|A| \neq 0$ → non singular Matrix

Inverse of a Matrix

A →

B →

If $\boxed{AB = BA = I}$

Properties of Inverse of a Matrix

Property 1

Inverse of only a non singular square matrix exists.

Property 2

If A is a non-singular matrix, then $|A^{-1}| = \frac{1}{|A|}$

Property 3

If A and B are invertible matrices of the same order, then $(AB)^{-1} = B^{-1}A^{-1}$.

Property 4

If A is an invertible matrix then a) $(A^{-1})^{-1} = A$

b) $(A^K)^{-1} = (A^{-1})^K = A^{-K}$, $K \in N$

Nature of Solutions (For Non Homogeneous System of Equations)

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned}$$

we can replace these 3 linear equations with the following matrix equation →

$$\underbrace{\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}}_{B}$$

$$X = \frac{(\text{adj } A)B}{|A|}$$

System of Linear Equation

Consistent System - the system of equation is said to be consistent if it has one or more than one solutions.

Inconsistent System - the system of equation is in-consistent if it has **no solution**

Nature of Solutions Non-Homogenous System of Equations

i) If $|A| \neq 0$ and $(\text{adj } A)B = 0$ then the system is consistent and has a unique trivial solution given by $x = 0$

ii) If $|A| \neq 0$ and $(\text{adj } A)B \neq 0$ then the system is consistent and has a unique Non trivial solution given by $x = A^{-1}B$.

iii) If $|A| = 0$ and $(\text{adj } A)B = 0$, then the system is consistent and has infinitely many solutions.

iv) If $|A| = 0$ and $(\text{adj } A)B \neq 0$, then the system is inconsis-tent having no solutions.

Nature of Solutions - Homogeneous system of Equations

b) For the homogenous system of equation $AX=0$

i) If $|A| \neq 0$, the solution is $x=0, y=0, z=0$. This is called the trivial solutions.

ii) If $|A|=0$, the system has infinitely many solutions. In such a case, we put one of the variable equal.